

2021-Ph-H1-Q13

Section: Particles and Waves

Topic: The Standard Model

Which statement about quarks is correct?

Solution:

- A proton = uud
 - A neutron = udd
 - Quarks have fractional charges: $\pm 1/3$ or $\pm 2/3$
- The correct charge combinations add to ± 1 or 0.

Answer: D

Guidance for Students:

Memorise the **quark structure** of protons and neutrons. Know the charges of up and down quarks.

Revision Tip:

Up = $+2/3$, Down = $-1/3$

Proton = uud $\rightarrow +1$

Neutron = udd $\rightarrow 0$